

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA1001	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	8-10-2026	Duration:	60 minutes

Code of Conduct

I P.SAI SEVITHA – 192525097 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P.Sai Sevitha

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Assess the Accessibility of Roads, Transport Facilities, and Essential Services

1. Identify the Project Vision and Stakeholders

Define the project's vision, objectives, and identify all key stakeholders involved.

Project Vision

To develop a system that assesses how easily people can access **roads, public transport facilities, hospitals, schools, markets, and other essential services** in a particular area.

Objectives

- Identify the accessibility of roads and transport facilities.
- Measure the distance and travel convenience to essential services.
- Identify areas with poor accessibility.
- Help authorities improve transportation and public services.
- Provide useful accessibility information to citizens.

Key Stakeholders

Stakeholder	Role
Citizens	Use the system to check accessibility
Government/Local Authorities	Plan infrastructure improvements
Transport Authorities	Improve public transport facilities
Healthcare Providers	Assess accessibility of hospitals
Educational Institutions	Assess accessibility of schools/colleges
Project Team	Develop and maintain the system

2. Create Epics and User Stories

Break down the requirements into Epics and formulate well-structured User Stories using the format:

As a <user>, I want <feature>, so that <benefit>.

Epic 1: Road Accessibility

User Story 1:

As a **citizen**, I want to **check road accessibility in my area**, so that **I can choose convenient routes**.

User Story 2:

As a **government authority**, I want to **identify areas with poor road connectivity**, so that **road infrastructure can be improved**.

Epic 2: Transport Accessibility

User Story 3:

As a **citizen**, I want to **find nearby bus stops and transport facilities**, so that **I can easily access public transportation**.

User Story 4:

As a **transport authority**, I want to **identify areas with limited public transport**, so that **new transport facilities can be planned**.

Epic 3: Essential Services Accessibility

User Story 5:

As a **citizen**, I want to **find nearby hospitals, schools, and markets**, so that **I can access essential services easily**.

User Story 6:

As a **government authority**, I want to **identify areas far from essential services**, so that **services can be improved in those areas**.

3. Define Acceptance Criteria

Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

User Story 1 – Road Accessibility

Given the user selects an area,
When the system analyzes the available roads,
Then the system should display the road accessibility level and available routes.

User Story 3 – Transport Accessibility

Given the user selects their location,
When the user searches for transport facilities,
Then the system should display nearby bus stops and other available transport facilities.

User Story 5 – Essential Services

Given the user enters a location,
When the user searches for essential services,
Then the system should display nearby hospitals, schools, markets, and their distance.

User Story 6 – Poor Accessibility Areas

Given accessibility data is available,
When the authority analyzes the area,
Then the system should identify areas having poor access to essential services.

4. Prioritize the Product Backlog

Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

Use **High, Medium, Low** priority.

Priority	User Story	Reason
High	Check road accessibility	Main project requirement
High	Find nearby transport facilities	Important for daily travel
High	Find essential services	Important for citizens
High	Identify poorly accessible areas	Helps authorities
Medium	Compare different areas	Useful for analysis
Medium	Generate accessibility report	Helps decision-making
Low	Additional visualization	Improves presentation
Low	Advanced filtering	Can be added later

Backlog Order

1. Road accessibility
2. Transport accessibility
3. Essential services accessibility
4. Identify poor-access areas
5. Area comparison
6. Accessibility reports
7. Advanced visualization

5. Plan the Sprint Backlog

Select high-priority user stories for the first sprint and divide them into actionable development tasks.

Sprint 1 Goal

Build the basic accessibility assessment system and provide users with information about **roads, transport facilities, and essential services.**

Selected User Stories

User Story	Priority
Check road accessibility	High
Find nearby transport facilities	High
Find nearby essential services	High

Development Tasks

Task 1 – Road Accessibility

- Collect road data.
- Process road information.
- Calculate accessibility.
- Display accessibility result.

Task 2 – Transport Facilities

- Collect bus stop/transport data.
- Identify nearby facilities.
- Display transport locations.

Task 3 – Essential Services

- Collect hospital, school and market data.
- Calculate distance.
- Display nearby services.

Task 4 – Testing

- Test location search.
- Test distance calculation.
- Test accessibility results.
- Fix errors.

6. Estimate Effort Using Story Points

Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

Using the **Fibonacci sequence: 1, 2, 3, 5, 8, 13**

User Story	Story Points	Justification
Check road accessibility	5	Requires road data processing and analysis
Find transport facilities	3	Requires searching and displaying locations
Find essential services	5	Requires multiple service categories and distance calculation
Identify poor-access areas	8	Requires data analysis and comparison
Generate accessibility report	3	Mainly involves presenting analyzed results

Why Fibonacci?

Story points increase as the **complexity, effort, and uncertainty** increase.

7. Sprint Goal and Expected Deliverables

Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Sprint Goal

To develop a basic system that assesses road accessibility and helps users locate nearby transport facilities and essential services.

Expected Deliverables

At the end of Sprint 1, the system should provide:

- 1. Road Accessibility Assessment**
 - Shows accessibility of selected roads/areas.
- 2. Transport Facility Search**
 - Shows nearby bus stops and transport facilities.
- 3. Essential Services Search**
 - Shows nearby hospitals, schools, and markets.
- 4. Distance Information**
 - Displays the approximate distance to facilities.
- 5. Basic Accessibility Result**
 - Identifies areas with good or poor accessibility.

Final Sprint Output

A working prototype for assessing accessibility of roads, transport facilities, and essential services.

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25